



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SUMATHI PETROL AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

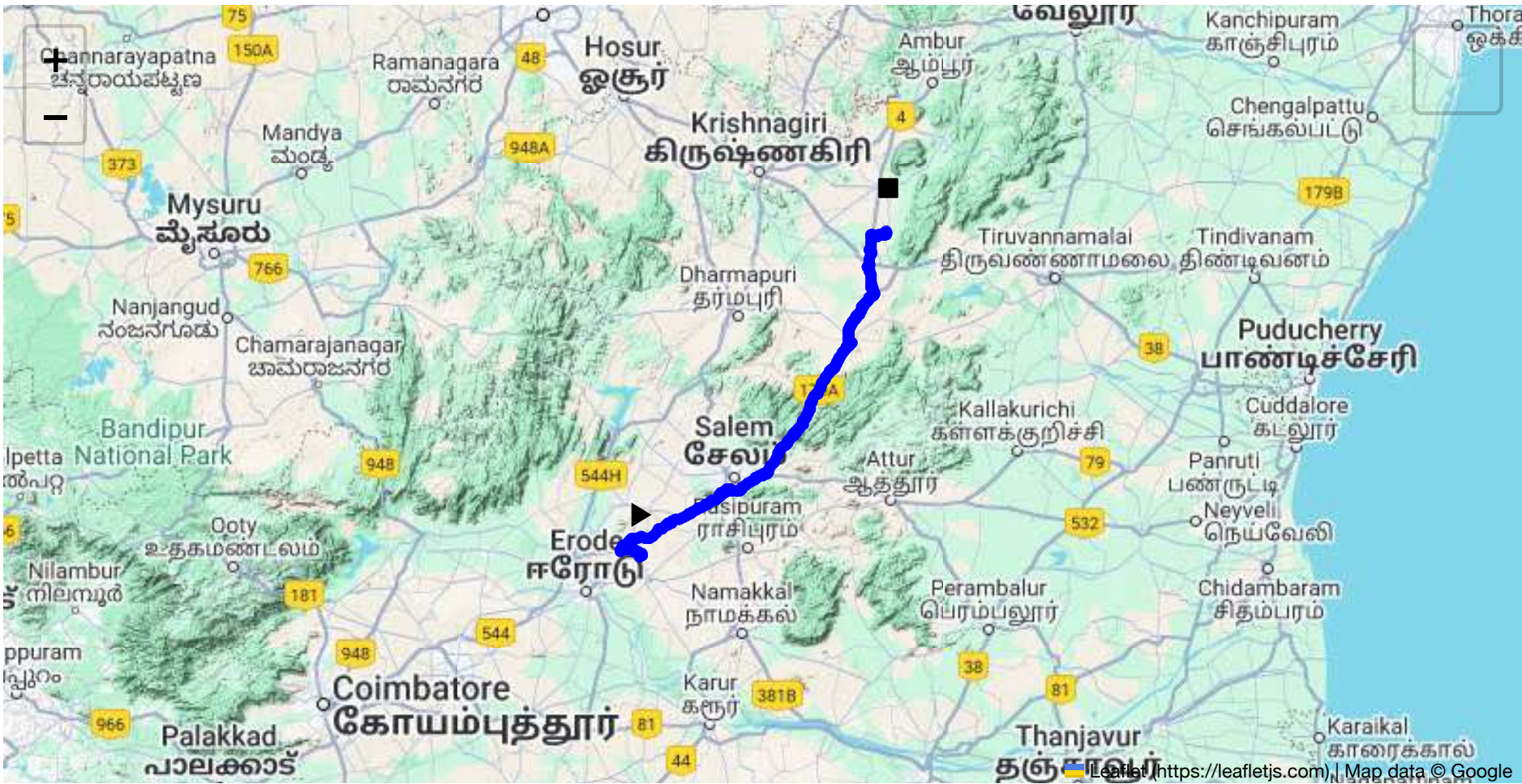
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 160.19 km
Estimated Duration: 3.7 hours
Adjusted Duration (Heavy Vehicle): 4.6 hours

Start: (11.4381, 77.8734)
End: (12.359412, 78.586237)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Perampatti spans approximately 160.19 kilometers, predominantly through rural and semi-urban areas of Tamil Nadu. It winds through several small towns and key junctions, connecting via Bhavani Main Road, Sankari Main Road, and METTUPATTI 4 ROAD, culminating in Perampatti. The route traverses a mix of national and state highways with varying traffic densities.

2. Typical Weather Conditions and Weather-related Hazards

Tamil Nadu generally experiences a tropical climate. The route may see:

- **Monsoon Rains (June to September):** Potential for heavy rainfall causing localized flooding and reduced visibility.
- **Summer Heat (March to May):** High temperatures may increase road surface tire wear and affect driver endurance.
- **Post-Monsoon (October to November):** Occasional thunderstorms could lead to slippery roads.

3. Traffic Patterns and Congestion-prone Areas

- **Peak Hours:** Morning (7:00-9:00 AM) and evening (5:00-7:00 PM) are typical peak hours.
- **Congestion-prone Areas:**
 - **Salem City Junction:** Notable for heavy traffic, especially during peak hours.
 - **Bhavani Main Road:** Can experience slow-moving traffic due to local commuting and market activities.

4. Road Quality and Infrastructure

- **National Highways:** Generally well-maintained with clear signage.
- **State Highways and Rural Roads:** Mixed quality; some roads may have potholes or uneven patches.
- **Bridges and Flyovers:** Adequately maintained, but periodic checks on load capacities are advisable for heavy vehicle routes.

5. Suggestions for Alternative Routes for Emergencies

- **Alternative 1:** Use Salem-Bangalore Highway for sections of the journey for faster transit if nearby routes are blocked.
- **Alternative 2:** Use local bypass roads around Salem for circumventing major traffic delays.

6. Local Regulations Affecting Hazardous Material Transport

- **Permits:** Required for different categories of hazardous materials.
- **Restricted Hours:** Transport of hazardous materials is typically restricted during peak daylight hours to minimize risks.
- **Safety Compliance:** Adherence to the Motor Vehicle Act and Central Motor Vehicle Rules is a must.

7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- **Salem District:** Has documented incidents involving chemical spills due to vehicular overturns, primarily due to speed or road negligence.
- **Recommendations:** Enhanced training for drivers on handling vehicle dynamics and adherence to speed limits.

8. Environmental Considerations and Sensitive Areas

- **Forested Zones:** Certain sections near Harur may affect local wildlife habitats.
- **Pollution Control:** Regions around Salem with higher population density necessitate strict anti-pollution protocols.

9. Analysis of Communication Coverage

- **Potential Dead Zones:** Certain rural stretches between Harur and Sukkampatti may experience weak mobile connectivity.
- **Recommendations:** Equip vehicles with satellite communication systems for uninterrupted connectivity.

10. Estimated Emergency Response Times

- **Urban Areas (e.g., Salem):** 15-20 minutes due to proximity of emergency services.
- **Rural Areas:** 30-45 minutes, factoring in distance and road conditions to emergency services.

12. Overall Summary of Risk Assessment

The route presents a moderate level of risk largely due to variable road conditions and traffic congestion near urban areas. Seasonal weather changes further add to potential hazards. It is critical to routinely update drivers on route conditions, equip them with emergency protocols, ensure vehicles meet all safety

compliance requirements, and possibly use technology aids like GPS and traffic updates for better route optimization. Regular communication with local authorities for updates on regulations and incidents will enhance transport safety.

Regular maintenance of vehicles, coupled with continuous driver training on defensive driving and hazard identification, will mitigate risks significantly.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	Medium	11.45353, 77.85697	30 KM/Hr
5	Turn	Medium	11.45781, 77.85203	30 KM/Hr
6	Turn	High	11.46307, 77.84941	15 KM/Hr
7	Turn	Medium	11.45546, 77.81800	30 KM/Hr
8	Blind Spot	Blind Spot	11.45605, 77.81683	10 KM/Hr
9	Turn	High	11.53578, 77.95802	15 KM/Hr
10	Blind Spot	Blind Spot	11.62145, 78.10740	10 KM/Hr
11	Turn	High	11.62106, 78.10767	15 KM/Hr
12	Blind Spot	Blind Spot	11.61257, 78.10446	10 KM/Hr
13	Turn	Medium	11.61243, 78.10512	30 KM/Hr
14	Turn	High	11.63306, 78.12722	15 KM/Hr
15	Turn	Medium	11.62851, 78.16509	30 KM/Hr
16	Turn	Medium	11.65782, 78.20430	30 KM/Hr
17	Turn	High	11.66387, 78.20665	15 KM/Hr
18	Turn	Medium	11.67131, 78.23138	30 KM/Hr
19	Turn	Medium	11.67433, 78.23905	30 KM/Hr
20	Turn	Medium	11.81219, 78.34077	30 KM/Hr
21	Turn	Medium	12.01548, 78.45650	30 KM/Hr
22	Turn	Medium	12.04309, 78.48361	30 KM/Hr
23	Turn	High	12.04490, 78.48355	15 KM/Hr
24	Turn	High	12.04984, 78.47480	15 KM/Hr
25	Turn	Medium	12.06014, 78.47787	30 KM/Hr
26	Turn	Medium	12.07207, 78.47727	30 KM/Hr
27	Turn	Medium	12.07420, 78.47634	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
28	Turn	Medium	12.17745, 78.54775	30 KM/Hr
29	Turn	Medium	12.17821, 78.54845	30 KM/Hr
30	Turn	High	12.18592, 78.54966	15 KM/Hr
31	Turn	Medium	12.18821, 78.54799	30 KM/Hr
32	Turn	Medium	12.20170, 78.54373	30 KM/Hr
33	Turn	Medium	12.24997, 78.53501	30 KM/Hr
34	Turn	Medium	12.25938, 78.53301	30 KM/Hr
35	Turn	Medium	12.30968, 78.53833	30 KM/Hr
36	Blind Spot	Blind Spot	12.34868, 78.54092	10 KM/Hr
37	Turn	High	12.34793, 78.55449	15 KM/Hr
38	Turn	High	12.34673, 78.55544	15 KM/Hr
39	Turn	Medium	12.34790, 78.55907	30 KM/Hr
40	Turn	Medium	12.34938, 78.56068	30 KM/Hr
41	Turn	High	12.34971, 78.56536	15 KM/Hr
42	Turn	Medium	12.35024, 78.56581	30 KM/Hr
43	Turn	High	12.35085, 78.56712	15 KM/Hr
44	Turn	Medium	12.34998, 78.56928	30 KM/Hr
45	Turn	Medium	12.35308, 78.57704	30 KM/Hr
46	Turn	Medium	12.35277, 78.57773	30 KM/Hr
47	Turn	Medium	12.35294, 78.57893	30 KM/Hr
48	Turn	High	12.35383, 78.57963	15 KM/Hr
49	Turn	High	12.35310, 78.58199	15 KM/Hr
50	Turn	High	12.35357, 78.58269	15 KM/Hr
51	Turn	High	12.35331, 78.58345	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
4	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
6	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
12	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
14	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
15	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
16	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
17	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
18	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
19	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
20	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
23	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
24	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium
25	hospital	Government Hospital Chinnankuppam	12.0110294, 78.4553224	30 km/h	Medium
32	hospital	Government Hospital, Harur	12.061178, 78.4756913	30 km/h	Medium
33	hospital	Subam Hospital	12.0595128, 78.4766713	30 km/h	Medium
34	hospital	Government Hospital, Uthangarai	12.26612, 78.535949	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
5	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
8	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
9	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
10	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
13	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
21	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
22	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium
26	school	Indian Matriculation Higher Secondary school	12.0061755, 78.4533388	30 km/h	Medium
27	school	Kurinji Matriculation Higher Secondary School	12.0156853, 78.4579056	30 km/h	Medium
28	school	Chinnan Kuppam Government High School	12.0190729, 78.4581955	30 km/h	Medium
29	college	Annai Arts and Science College	12.0217961, 78.462795	30 km/h	Medium
30	school	Harur International School (CBSE)	12.0299967, 78.4693492	30 km/h	Medium
31	school	Government Girls Hr Sec School	12.0488709, 78.4762447	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



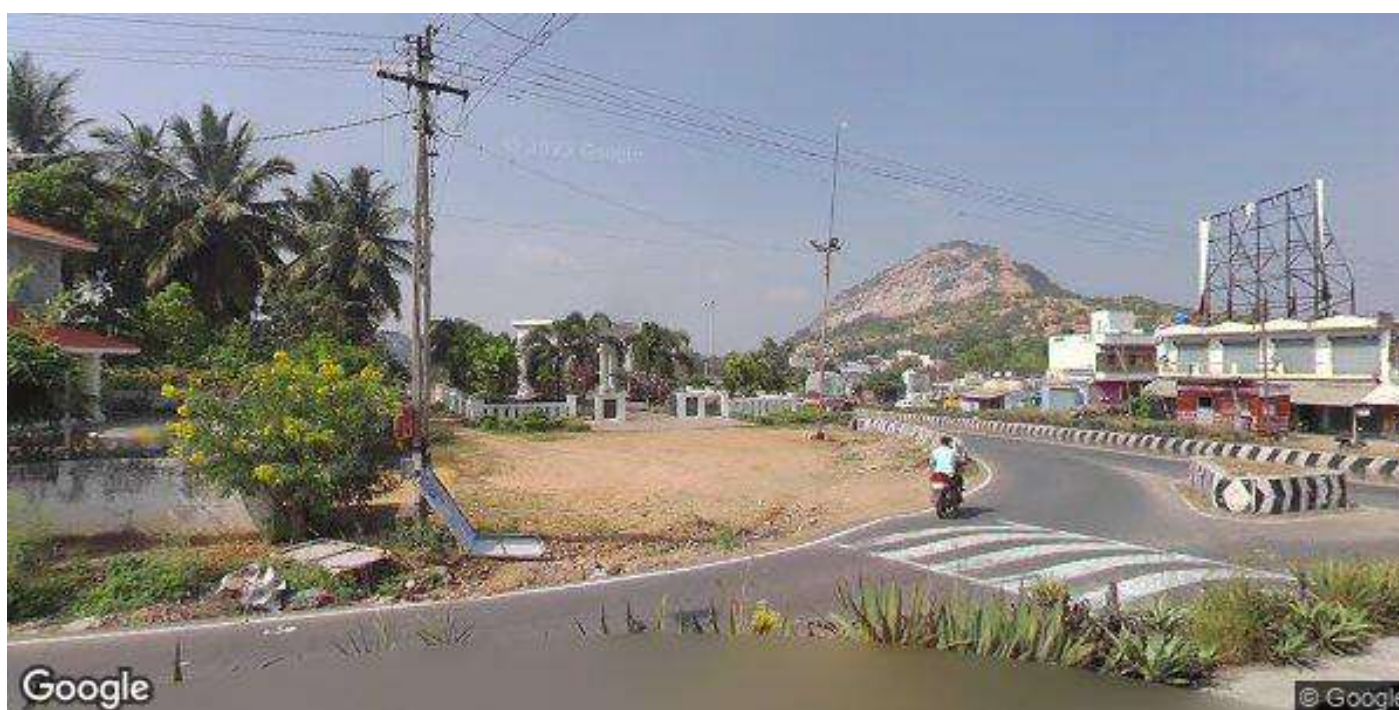
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45546, 77.81800



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.53578, 77.95802



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62145, 78.10740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.62106, 78.10767



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61257, 78.10446



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61243, 78.10512



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63306, 78.12722



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62851, 78.16509



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65782, 78.20430



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66387, 78.20665



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67131, 78.23138



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67433, 78.23905



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81219, 78.34077



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01548, 78.45650



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04309, 78.48361



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.04490, 78.48355



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.04984, 78.47480



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.06014, 78.47787



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.07207, 78.47727



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.07420, 78.47634



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17745, 78.54775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17821, 78.54845



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.18592, 78.54966



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.18821, 78.54799



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.20170, 78.54373



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24997, 78.53501



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25938, 78.53301



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.30968, 78.53833



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.34868, 78.54092



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.35331, 78.58345

